

Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Let

$$S(N, m) := Nm + \sum_{k=0}^{N-1} k$$

Let a be a positive integer. Determine the remainder when

$$\left(\sum_{i=1}^a S(a, i) - a \right)$$

is divided by 6.

$$\begin{aligned} & \sum_{i=1}^a S(a, i) - a \\ &= \sum_{i=1}^a \left(ai + \sum_{k=0}^{a-1} k \right) - a \\ &= \frac{a^2(a+1)}{2} + \frac{a^2(a-1)}{2} - a \\ &= a^3 - a \\ &= (a-1)a(a+1) \end{aligned}$$

Note that this is the product of 3 consecutive integers, which is always divisible by 6. Hence the remainder when the expression is divided by 6 is $\boxed{0}$